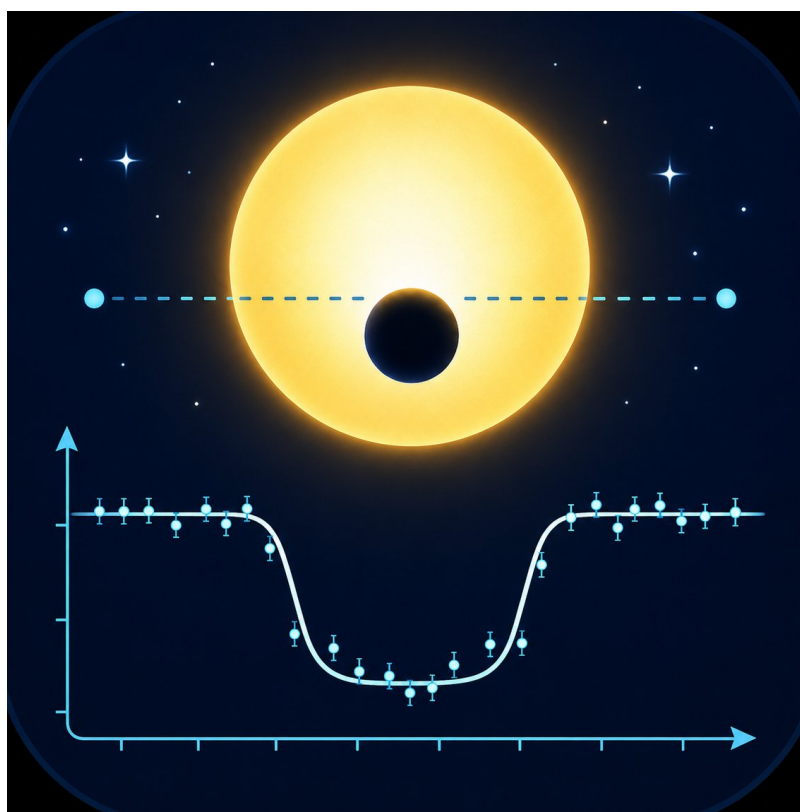


ExoPhotoCurve 1.0.0 User Manual

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ExoPhotoCurve — User Guide

ExoPhotoCurve is a program designed to create, inspect, correct, and analyze photometric light curves, with a special focus on exoplanet transits.

The program can be used in two main ways:

1. **Analyze an existing light curve**, for example one exported from AstrolImageJ, HOPS, or another photometry program.
2. **Create a light curve from calibrated and aligned FITS images**, using the integrated aperture photometry tool.

ExoPhotoCurve does not replace image calibration software. Bias, dark, and flat-field correction, image alignment, and plate solving must be done before using this program. ExoPhotoCurve works with already prepared light curves or with already calibrated and aligned images.

Purpose of the program

The goal of ExoPhotoCurve is to let the user go from calibrated images or from a photometric table to a corrected, modeled, and documented light curve.

The program helps the user control:

comparison stars
rejected points
detrending
timing
transit model
light-curve quality
final report

In this way, the analysis becomes more reproducible and less dependent on undocumented manual trial and error.

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1. Starting the program

Start ExoPhotoCurve by running the main Python script or the Windows executable.

The main window contains:

- a control panel on the left;
- the light-curve plot on the right;
- several working tabs: Data, Comp stars, Cleaning, Detrend, Transit modeling, Statistics, Plot, Style, and Binning.

If you already have a light curve, you can load it directly.
If you want to create a light curve from FITS images, use:

Build light curve

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2. Creating a light curve from FITS images

The **Build light curve** button opens the independent aperture photometry tool.

This tool creates a photometric table compatible with the main ExoPhotoCurve pipeline. The output table uses column names similar to those produced by AstroImageJ, so it can be loaded and analyzed directly by the main program.

2.1 Image requirements

The FITS images must already be:

- bias/dark/flat calibrated;
- aligned;
- preferably all the same size;
- correctly ordered in time;
- equipped with reliable timing information in the FITS header.

The tool does not perform calibration, image alignment, or plate solving.

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2.2 Loading the FITS sequence

In the Aperture photometry window, select the folder containing your FITS images.

The program loads the sequence and displays a reference image. You can move through the sequence using:

arrow keys
previous/next buttons
Shift + mouse wheel

You can inspect the image using zoom and pan:

mouse wheel = zoom
click + drag = pan
Fit view = show full image

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2.3 Choosing the apertures

Set the aperture parameters:

aperture radius
sky inner radius

sky outer radius
search radius

The inner circle is the stellar aperture.

The outer annulus is used to estimate the sky background.

You can change the aperture sizes at any time until you close the window, even after selecting the target and comparison stars. The new aperture sizes are calculated automatically and shown in the image window.

When you move the mouse over the image, the program shows the aperture at the current position. This helps you check whether the selected radius is appropriate for the stars. The aperture is color-coded according to the peak luminosity measured in the inner circle: yellow means that the selected star has low luminosity and probably a low signal-to-noise ratio. Green means that the selected star is within the optimal luminosity range, which is set by default between 0.25 and 0.66 of the dynamic range of a standard 16-bit FITS image (roughly between 18,000 and 35,000 ADU). You can change this range in the “OK frac” field. Red means that the star is approaching saturation and probably falls within the nonlinear range. Always avoid red apertures!

This color feedback is based only on the luminosity of the stars in the selected image. After selecting the apertures, use the “Check peak” button. By default, it measures the peak luminosity in a subsample of 15 images and updates the aperture colors if at least one aperture falls outside the optimal range in at least one image.

The program also displays useful information about the region under the cursor, such as the pixel value, peak value, and mean signal inside the aperture.

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2.4 Selecting the target and comparison stars

Before clicking on the image, choose which type of star you want to add:

Target
Comparison
Check

Then click near the star.

If **Snap click** is enabled, the program automatically moves the aperture to the local centroid of the nearest star within the search radius.

This means you do not need to click exactly at the center of the star.

The selected apertures remain visible on the image.

You can delete or modify apertures before running the photometry.

You can also try automatic comparison-star selection by pressing the “Auto find comps” button. After you have manually selected the target, this option selects comparison stars that meet the following constraints: peak luminosity in the green range and no contamination from nearby stars.

Whether you select comparison stars manually or automatically, it is advisable to start with a broad range of suitable stars. A finer selection will be made later in the main program, under the “Comp stars” tab.

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2.5 Saving and loading apertures

You can save the aperture set to an external file.

This is useful if you want to repeat the photometry on the same sequence, or if you want to change only a few parameters without selecting all stars again.

When you reload an aperture set, the program restores the saved positions and displays them on the image.

The saved coordinates are reference positions. If Recenter frames is enabled, the program recalculates the centroid of each star in every image during photometry, within the selected search radius.

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2.6 Recentering apertures during photometry

The option:

Recenter frames

makes the program search for the centroid of each star in every image.

This compensates for small alignment errors or small drifts of the field.

If the images are perfectly aligned, you can disable this option, but in most cases it is safer to keep it enabled.

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2.7 Timing information in FITS images

Timing is one of the most important aspects of transit photometry.

Many FITS files store DATE-OBS or JD.UTC as the **start of the exposure**, not the mid-exposure time.

For 300-second exposures, the difference between exposure start and mid-exposure is:

150 seconds = 2.5 minutes

This can produce a visible offset in the measured O-C.

In the photometry window, choose:

FITS time ref:

- Exposure start
- Mid-exposure
- Exposure end

Then choose which time should be exported as the main JD.UTC column:

Main JD.UTC:
- Header time
- Mid-exposure corrected

A safe workflow is:

FITS time ref = Exposure start
Main JD.UTC = Mid-exposure corrected

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2.8 Running the photometry

When the apertures are ready, click one of these buttons:

Run photometry
Run + load in main

The first option runs the photometry only. The second option runs the photometry, closes the window, and automatically loads the raw photometry file in the main program. The program measures the fluxes on the full FITS sequence and creates a photometric table.

The output file can contain columns such as:

JD.UTC
JD.UTC_header
JD.UTC_start
JD.UTC_mid
JD.UTC_end
AIRMASS
EXPTIME
FILTER
Source-Sky_T1
Source_Error_T1
Source-Sky_C2
Source_Error_C2
Source-Sky_C3
Source_Error_C3
rel_flux_T1
rel_flux_err_T1

From this point on, the light curve is analyzed using the standard ExoPhotoCurve pipeline.

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3. Loading an existing light curve

If you already have a light curve exported from AstrolImageJ, ExoClock, or ExoPhotoCurve, you can load it directly into the main window.

Click:

Browse
Load table

The program reads the table and tries to identify the main columns automatically.

For AstrolmageJ files, common columns are:

JD.UTC
JD.UTC_B
BJD.TDB
rel_flux_T1_dfn
rel_flux_err_T1_dfn
rel_flux_T1_dfn_model
rel_flux_T1_dfn_residual

If some columns are not recognized correctly, you can select them manually in the Data tab. In general, the photometry file should use column names without special characters, such as #.

4. Data tab

The **Data** tab is used to choose which columns are used for plotting and analysis.

The main columns are:

X / time
Light curve
Light-curve error
Model
Residuals
Residual errors

If your file does not contain a model or residuals, leave those fields as:

-- None --

The model can be generated later in the **Transit** tab.

5. Comp stars tab

The Comp stars tab is used to select the best comparison stars when the loaded file contains individual stellar fluxes. If you created the raw light curve with the Build light curve tool, use this tab to check and select the final set of comparison stars from the group selected during photometry. This is a very important step: in most cases, you need to inspect the comparison stars carefully and remove those that are unsuitable because they are variable or unstable over the sequence. You will rarely end up with the same number of comparison stars that you selected during photometry.

This applies to complete AstrolmageJ tables and to files generated by the ExoPhotoCurve aperture photometry tool.

The tab becomes active only if the program detects columns such as:

Source-Sky_T1
Source-Sky_C2
Source-Sky_C3

Source_Error_T1
Source_Error_C2
Source_Error_C3

If these columns are not present, the tab displays a warning and cannot be used.

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5.1 Manual comparison-star selection

The list shows the available comparison stars.

You can activate or deactivate each star by clicking on its name.

You can also view the light curve of any comparison star by activating the “Plot selected comp” option.

When the subset changes, the light curve is recalculated automatically.

This allows you to quickly check the effect of each comparison star on the final light curve.

Suggested workflow:

- Start by looking at each check star individually with the “Plot selected comp” option enabled, and deselect stars with clearly variable or nonlinear behavior. For now, keep stars that show only a linear trend.
- Continue the fine selection by observing the target light curve with “Plot selected comp” disabled while activating and deactivating one comparison star at a time. If the curve improves when you deselect a comparison star, leave it deselected and move to the next comparison star until you reach the end of the list. Repeat this step from the first active comparison star, considering only the comparison stars that are still active. For a reliable light curve, at least three comparison stars should be used. Using more than 10-15 comparison stars is usually not useful.

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5.2 Automatic comparison-star optimization

The program can automatically search for the subset of comparison stars that produces the cleanest light curve. This is an experimental feature and should be used with extreme caution.

You can use two workflows:

manual selection → automatic optimization
automatic optimization → manual refinement

In the first case, you manually exclude poor comparison stars and then let the optimizer work only on the remaining active stars.

In the second case, the program selects a subset automatically and you refine it manually afterwards.

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5.3 Sending the optimized curve to the pipeline

If:

Send optimized curve to Data/Transit

is enabled, the optimized curve is automatically selected as the main light curve in the Data tab.

You can then proceed with cleaning, detrending and transit fitting.

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6. Cleaning tab

The **Cleaning** tab is used to remove problematic points.

There are two main methods:

automatic sigma clipping
manual point rejection

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6.1 Sigma clipping

Sigma clipping identifies outliers based on their distance from the curve or from the residuals.

For exoplanet transits, use this function carefully, because overly aggressive clipping may remove real transit points.

A reasonable starting point is:

Target: Residuals
Sigma threshold: 3
Center: Median
Scale: MAD

For more accurate sigma clipping, consider fitting the transit first in the “Transit modeling” tab.

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6.2 Manual point rejection

You can enable manual point editing.

When it is active, clicking near a point in the plot excludes or restores that point.

Manually rejected points are removed from the plot, statistics and transit fit, unless you explicitly choose to export rejected points.

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7. Detrend tab

The **Detrend** tab is used to correct systematic trends in the light curve.

You can use one or more regressors at the same time, as in AstrolmageJ.

Examples of possible regressors are:

JD.UTC / time
AIRMASS
FWHM
Sky background
X/Y centroid
Peak
Meridian flip

The detrending is applied multiplicatively:

corrected flux = observed flux / baseline

This is appropriate for relative-flux light curves.

For better results, consider fitting the transit first. Then activate the “Consider fit model” option, which will give you better results with any detrending method, especially with the meridian flip.

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7.1 Choosing regressors

By default, no regressor is active.

You can select regressors manually, or press Suggested to let the program enable likely useful regressors.

You can use several regressors together, for example:

JD.UTC + AIRMASS

or:

JD.UTC + AIRMASS + FWHM

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7.2 Masking the expected transit during detrending

For exoplanet transits, it is usually recommended to use:

Mask expected transit

This tells the program not to use in-transit points when estimating the baseline.

This reduces the risk that the detrending will distort the transit depth.

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7.3 Meridian flip

If the telescope performed a meridian flip during the sequence, you can use the dedicated correction.

A meridian flip may introduce a jump in flux level or a change in trend, because the field moves to a different region of the detector.

In the **Detrend** tab, enable:

Meridian flip

and enter only the decimal part of the flip time, as in AstrolmageJ.

Example:

JD.UTC flip = 2461203.771

In the program, enter:

.771

You can press:

Update line

to display a vertical line in the plot at the flip position. This is only a visual check.

To actually apply the correction, click:

Run detrending

Available modes are:

Step only

Step + after-flip slope

Robust level match

The recommended mode is **Robust level match**, because it corrects a flux-level jump without adding too much freedom.

Use **Step + after-flip slope** only if there is a clear change in slope after the flip.

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7.4 Removing detrending

If you want to return to the curve before detrending, click:

Clear detrending

This does not reset the full program. It removes only the detrending and returns to the source curve.

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8. Transit modeling tab

The **Transit modeling** tab is used to model an exoplanet transit and produce a diagnostic report.

The typical workflow is:

1. Select the planet
2. Select the filter
3. Enter the exposure time
4. Select the input time system
5. Select whether the time stamps correspond to exposure start, mid-exposure, or exposure end
6. Enter the observatory coordinates
7. Click Run transit model

8.1 Planet catalogs

ExoPhotoCurve can use different catalogs:

NASA
ExoClock
custom catalog

You can quickly switch between NASA and ExoClock using the dedicated buttons.

The ExoClock catalog is often useful for updated ephemerides.

The NASA catalog can provide geometric parameters useful for the physical transit model.

You can also load a custom catalog, as long as it uses a compatible column format.

8.2 Automatic planet detection

When a file is loaded, ExoPhotoCurve tries to detect the planet from the filename.

Examples:

hats24b_20260606_lightcurve.txt → HATS-24b
wasp-15b_lightcurve.txt → WASP-15b

Always check that the selected planet is correct before running the fit.

8.3 Photometric filter and limb darkening

The selected filter is used by the physical model to choose approximate limb-darkening coefficients.

Changing the filter should not drastically change the transit, but it can slightly affect:

transit shape
ingress and egress

estimated Rp/Rs
residuals

The report shows the filter, limb-darkening law, and coefficients used.

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8.4 Time system and BJD_TDB correction

Transit timing requires careful time handling.

ExoPhotoCurve can convert JD_UTC times to BJD_TDB using:

target coordinates
observatory coordinates
exposure time
time-stamp reference

If the time column is JD_UTC, select:

Input time system = JD_UTC

If the times are already in BJD_TDB, select:

Input time system = BJD_TDB

The field:

Time stamp reference

defines whether each point represents:

Exposure start
Mid-exposure
Exposure end

This choice is very important. With 300-second exposures, confusing exposure start with mid-exposure produces a 2.5-minute timing error.

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8.5 Fit baseline

In the **Transit** tab, you can choose a baseline model:

Constant
Linear
Quadratic

This baseline corrects small remaining trends during the transit fit.

If you have already applied a good detrending in the **Detrend** tab, **Constant** is often sufficient.

If residual trends remain, try **Linear** or **Quadratic**.

Use the quadratic baseline with caution if there is limited out-of-transit coverage. Usually, Quadratic will overestimate the transit depth, especially if you have already performed detrending in the “Detrend” tab.

8.6 Detrended or raw display

ExoPhotoCurve can display the transit fit in two ways:

Detrended flux
Raw flux with baseline

The recommended mode for final plots is:

Detrended flux

In this mode, the program shows the baseline-corrected light curve and clean transit models.

The mode:

Raw flux with baseline

is useful for checking how the baseline fits the original data.

8.7 Timing markers in the plot

You can show timing labels in the plot:

Predicted times
Calculated times

If **Predicted times** is enabled, the plot shows:

Predicted start
Predicted Tmid
Predicted end

If **Calculated times** is enabled, the plot shows:

Calculated start
Calculated Tmid
Calculated end

If both are enabled, the program also displays the O-C between the predicted and measured mid-transit times.

8.8 Tmid ref

The field:

Tmid ref

allows you to manually enter a predicted mid-transit time in BJD_TDB.

Use this only if you have a predicted mid-time for the specific transit observed.

Do not enter an old ephemeris T0 here unless it corresponds to the observed transit.

9. Transit report

After **Run transit model**, ExoPhotoCurve produces a report containing:

- planet
- filter
- exposure time
- time system
- BJD_TDB correction
- target coordinates
- observatory coordinates
- catalog used
- T0 and period
- predicted Tmid
- observed Tmid
- O-C
- Rp/Rs
- depth
- duration
- residual RMS
- transit SNR
- autocorrelation
- diagnostic quality flags

The O-C is defined as:

$$\text{O-C} = \text{observed Tmid} - \text{predicted Tmid}$$

A positive O-C means that the transit occurred later than predicted.

A negative O-C means that it occurred earlier than predicted.

10. Reproducibility recipe

ExoPhotoCurve also saves a complete recipe of the analysis.

The recipe includes:

- loaded file
- catalog used
- selected columns
- active comparison stars
- rejected points
- sigma clipping settings
- detrending regressors
- meridian flip settings
- fit baseline
- filter
- time-stamp settings
- observatory coordinates
- main final parameters

You can save:

Save model results

to save the complete diagnostic report, or:

Save recipe

to save only the operational recipe.

This makes the analysis reproducible.

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11. Saving the final light curve

The button:

Save curve

exports the light curve in its current analysis state.

The file can include:

JD.UTC
time_input
time_bjd_tdb
time
flux
flux_error
model
residual
baseline
point masks

To load the result into ExoClock, usually use:

JD.UTC
flux
flux_error

If you manually rejected points, ExoPhotoCurve saves only the points used in the analysis by default.

Enable:

Export rejected points

if you want to save rejected points as well, together with mask columns.

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12. Binning

The **Binning** tab allows you to overplot binned points on the original light curve.

Binning is useful for visualizing the general shape of the transit, but it should not replace the analysis of the original data points.

You can choose the number of points per bin and the style of the binned points.

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13. Statistics

The **Statistics** tab computes basic quantities for the selected light curve, such as:

number of points
time span
median cadence
mean and median flux
RMS
amplitude
median error
RMS / median error

This is useful for quickly checking the quality of the light curve.

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14. Reset and useful controls

ExoPhotoCurve includes several controls for reverting changes:

Reset view/data

restores the loaded table and removes temporary products, fits, binning, clipping, and generated columns.

Clear detrending

removes only the detrending, while keeping the rest of the setup.

Show transit fit on plot

shows or hides the transit model in the plot.

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15. Recommended workflow for an existing light curve

1. Load table
2. Check the columns in the Data tab
3. Use Comp stars if the file contains individual stellar fluxes
4. Remove problematic points in Cleaning
5. Apply detrending if needed
6. Select planet, filter, exposure time and time-stamp reference in Transit
7. Run transit model
8. Check plot, residuals, RMS and O-C
9. Save model results, Save recipe and Save curve

.

16. Recommended workflow starting from FITS images

If you want to do everything inside ExoPhotoCurve:

1. Build light curve
2. Load the FITS folder
3. Inspect image, zoom, and stretch
4. Set aperture and sky annulus
5. Select target and comparison stars
6. Check FITS time ref and Main JD_UTC
7. Run photometry
8. Save the photometric table
9. Load the table into the main window
10. Use Comp stars to optimize the comparison subset
11. Apply manual cleaning or sigma clipping
12. Apply detrending with useful regressors
13. Run transit model
14. Save model results, recipe, and final curve

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17. Practical advice

For reliable results:

- use well-calibrated images;
- avoid saturated stars;
- choose stable comparison stars that are not too faint;
- always check the time system;
- verify whether the FITS time is exposure start or mid-exposure;
- use meridian flip correction only when needed;
- avoid overly aggressive detrending if the transit is poorly covered;
- always inspect the residuals, not only the fitted curve;
- save the analysis recipe.

A good fit is not only a model passing through the data. A good analysis should have small residuals, no obvious trends, consistent timing, and a reproducible procedure.

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18. Program limitations

ExoPhotoCurve is designed for differential photometry, diagnostics, and operational/preliminary exoplanet transit analysis.

It does not perform:

bias/dark/flat calibration
image alignment
plate solving
PSF photometry

full MCMC analysis
advanced physical modeling

The program is designed to be simple, transparent, and practical. For final professional analyses, the results can be used as a basis or comparison, but they should be interpreted carefully.

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